

9708/32/o/n/19

1-B:

Total costs incurred by the oil companies in *these circumstances whereby they are obligated to repair the damage they do to the environment* represents "social cost"

- ➔ Social Cost is the sum of private cost and external costs.
- ➔ The oil company will incur **private costs**(in terms of the costs of rent, wages, interest payments, and other variable and fixed costs incurred in operating an oil extraction business.)
- ➔ And also, they are required to pay for the **external costs** (by paying for the damage they do the environment).
- ➔ Since, they pay for both: the external costs and the private costs, this is why "key B", becomes correct.

2-C:

The statement "no one can be made better off without making someone else worse off" is known as the "Pareto optimality" criterion. It means that resources are allocated efficiently, but it doesn't necessarily imply that the distribution of income is fair or socially acceptable.

It occurs when both: productive and allocative efficiency is achieved.

A) Allocative efficiency: If no one can be made better off without making someone else worse off, it means that resources are being used in the most efficient way possible according to the society, meeting the conditions for allocative efficiency.

B) Productive efficiency: Pareto optimality also implies that resources are being used in the most efficient way possible, meeting the conditions for productive efficiency.

D) Operating on the production possibility curve: If resources are being used efficiently, the economy must be operating on its production possibility curve, which represents the maximum output possible given the available resources.

This explains why option C will be correct.

3-B:

Since, the road building scheme is “socially beneficial”, this means that $SB > SC$.

- The **private costs** are of \$800m, and the **external costs** are of \$150m, which means that the **social costs are of \$950m**.
- The **private benefits** are \$800m, and we are currently not being given information about the external benefits, which means that the expression for “social benefit” must be “ $800+x$ ” (let X be the external benefit)
- Since, the **social costs are 950m**, we need **social benefits of greater than 950** (for SB to be greater than SC , and for the project to be “socially beneficial”). This means that **the “external benefit” that is the missing value must be greater than “150”**

A – External benefits(of greater than 150) are not equal to the private benefits(of 800)

B – **External benefits should exceed external costs**(which are 150), to make the project “socially beneficial”. This can be seen above, which is why, this becomes correct.

C – External benefits should not exceed \$200m, instead, **they should exceed \$150m**

D – This is incorrect because if there were no external benefits, then the project would not have been socially beneficial.

4-C:

When a consumer's marginal utility (the additional satisfaction or value they get from consuming an additional unit of good), **which also represents their willingness to pay** is greater than the **price they pay**, this means they are getting more value than they are paying for. **This difference between the utility and the price is called consumer surplus.**

This explains why key C is correct.

A) A Giffen good is a rare type of good where the demand increases when the price rises, defying the law of demand. This occurs when the good is a staple and the consumer has no close substitutes.

B) An inferior good is a good whose demand decreases when income rises. It is a good that people tend to consume less of as their income increases.

D) Producer surplus is the difference between the price a producer receives and the minimum price they are willing to accept.

5-B:

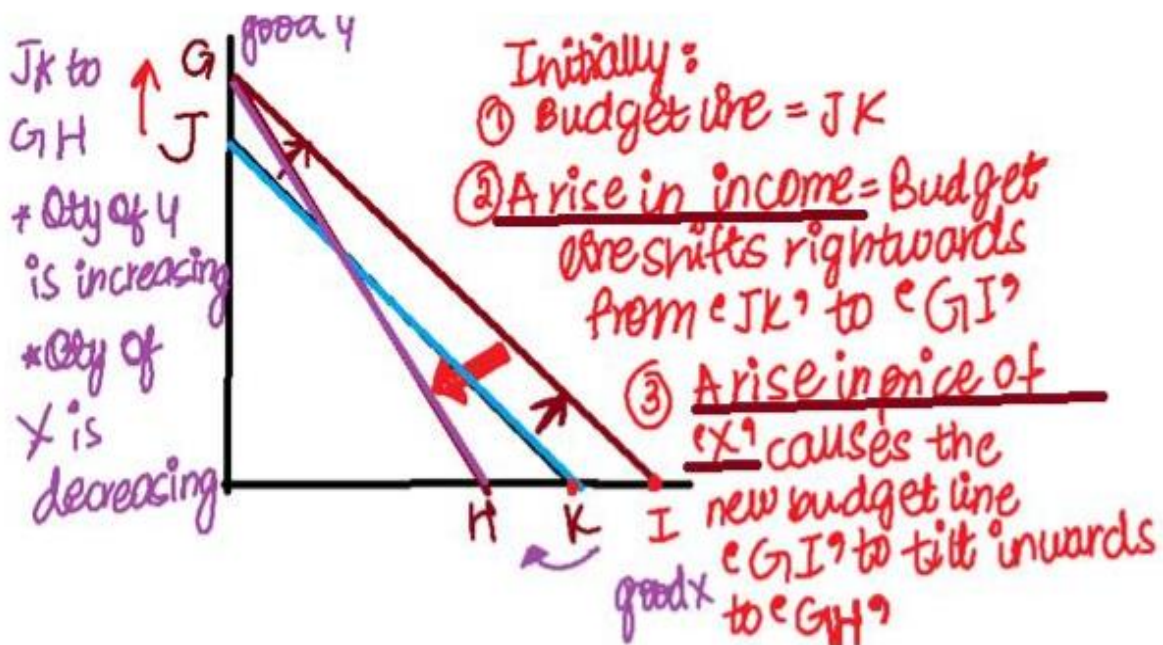
The table below shows the units of additional satisfaction she gets from consuming each bar.

1 st chocolate bar 16	2 nd chocolate bar 13	3 rd chocolate bar 10
1 st toffee bar 14	2 nd toffee bar 12	3 rd toffee bar 9

What should she buy to maximise her total utility?

- A 1 chocolate bar and 2 toffee bars $\Rightarrow 16 + 14 + 12 = 42$
- B 2 chocolate bars and 1 toffee bar $\Rightarrow 16 + 13 + 14 = 43$ → highest utility**
- C 3 chocolate bars $\Rightarrow 16 + 13 + 10 = 39$
- D 3 toffee bars $\Rightarrow 14 + 12 + 9 = 35$

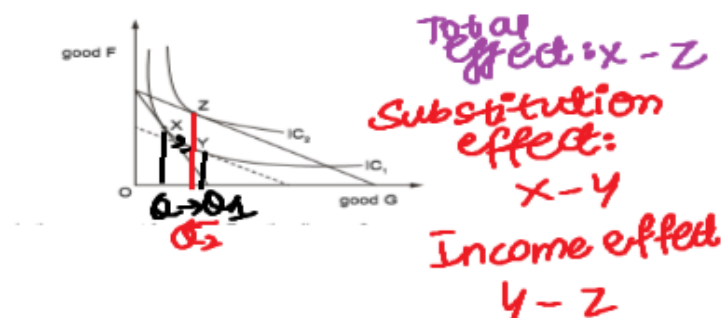
6-A:



The diagram above explains that the income of consumer has risen, which has led to an increase in qty purchased of good Y, and also that the price of good X has risen, which has led to a contraction in qty purchased of X.

This is why, key A, becomes correct.

7-A:



A – Since the income effect and the substitution effect (represented through the movement along the indifference curve) move in the opposite direction, the above good is said to be an “inferior good”

B – In case of a Giffen good, the income effect outweighs the substitution effect, which is why, this becomes incorrect

C – This is incorrect because in case of a normal good, the income effect and the substitution effect move in the same direction

D – In case of a Giffen good, the income effect outweighs the substitution effect, which is why, this becomes incorrect

8-C: This explains why key C is correct.

output (units)	total cost (\$)	total revenue (\$)	Profit	ATC
1000	8000	10000	2000	8
2000	12000	18000	6000	6
3000	19000	24000	5000	6.3
4000	23000	28000	5000	5.75
5000	25000	25000	0	5

Handwritten notes and arrows:

- TC/Qty (Total Cost per unit) points to the ATC column.
- Highest profit (option A) is achieved at 2000 units. (Arrow points to 6000 profit at 2000 units)
- Lowest average cost at 5000 units (option D) (Arrow points to 5 ATC at 5000 units)
- * Sales (option B) refers to the volume or qty sold. It is the maximum at 5000 units (Arrow points to 5000 output)
- * Sales revenue (option C) is the highest at 4000 units. (Arrow points to 28000 total revenue at 4000 units)

9-D:

A) This is incorrect. The fact that commercial banks link their interest rates to that of the central bank **does not imply perfect competition**. This is because, they do not offer **homogeneous or standardized products or services**.

B) This is also incorrect, because banks engage in **product differentiation** : varying interest rates, and other unique features. Perfect competition **requires identical products**, which is not the case in the banking industry.

C) While commercial banks do face competition, their market structure is better characterized as an oligopoly, as they are **“dominated by a few large banks** that have **significant market power and influence.”** Although, there may be many small banks that exist, **but there are only a few large firms (with numerous branches, and in turn substantial economies of scale)**, that have the **highest market share**. This means that, they are not a part of **“monopolistic competition”**

D) Oligopoly, because they are affected by the actions of other banks: This is the correct answer. **In an oligopoly, there are only a few large firms that dominate the market, and they have a substantial impact on each other's actions and decisions.** Commercial banks fit this description because **their behavior is “interdependent” , and they closely monitor and respond to the actions of other banks. Decisions such as interest rates, fees, and lending policies by one bank can influence the decisions of other banks in the industry.**

10-D:

D - A cartel is a group of firms that collude to restrict output and raise prices. For a cartel to be successful, it needs to have a mechanism to enforce production quotas among its members **so that existing market shares are maintained**. This means that each firm in the cartel must agree to produce a certain quantity, and the cartel must have a way to monitor and enforce compliance with these quotas. **Without enforced quotas, cartel members would have an incentive to cheat and produce more than their agreed-upon share, which would undermine the cartel's effectiveness.**

A) A large number of firms in the industry would actually make it harder for the cartel to coordinate and enforce production quotas. For a cartel to be effective, there must be a small number of large, competing players in the industry.

B) If each firm has a differentiated product, this would reduce the incentive for firms to collude, as they would have less to gain from joint profit-maximizing behavior.

C) Low barriers of entry to the industry would make it easier for new firms to enter the industry, which would increase competition and make it harder for the cartel to maintain its market power.

11-B:

B - A contestable market is a market where firms can freely enter and exit, and there are no or very less barriers to entry or exit. This means that if a firm is earning abnormal profits, new firms can enter the market to compete away those profits. Similarly, if a firm is making losses, it can easily exit the market without incurring significant costs.

A – It is not necessary for the market to be supplied by a large number of competing firms for the market to be contestable. Even a monopoly can face a contestable situation, that is when, a monopoly being a monopoly cannot act as a monopoly due to the potential threat of competition.

C – Even in a contestable market, firms can earn supernormal profits in the short-run. The only thing that will happen here is that if a firm is earning abnormal profits, new firms can enter the market to compete away those profits. **This means that, they won't be able to earn supernormal profits in the long-run.**

D – Product differentiation is not essential for a market to be contestable. Instead the need to differentiate a product from rivals will act as a barrier to entry, which will make the market less contestable.

12-A:

The correct answer is A) Directors prefer company growth to greater shareholder dividends.

The principal-agent problem refers to a situation where there is a conflict of interest between two parties, typically a principal (shareholders) and an agent (directors or managers). In a company, shareholders (principals) may want the company to be run in a way that maximizes their profits (dividends), while directors or managers (agents) may have different goals, such as maximizing revenue from sales because their salaries and bonuses are linked to sales revenue. When managers ignore worker's concerns, and workers go on a strike against managers' re-organization plans, this represents a labor dispute, which is not related to the principal-agent problem between shareholders and directors.

Similarly, if shareholders participate in day to day activities, that is, they involve themselves in determining the prices of products, this will not result in principal-agent problem.

13-B:

The table shows the output of chairs at a factory when different numbers of workers are employed.

number of workers	0	1	2	3	4	5
number of chairs produced	0	7	17	26	34	40
marginal product of labour	0	7	10	<u>9</u>	8	6

** Marginal product of labour diminishes when three workers are hired.*

Diminishing returns to labor sets in when the marginal product of labor diminishes, and that occurs **when the third worker is employed.**

14-D:

Here, an upward trend has been shown in the wage rates. This represents that either the demand for labor has increased or the supply of labor has fallen.

A – An increase in capital intensive production represents a fall in the demand for labor, which will bring a downward pressure on the wage rate.

B – An increase in the number of industrial workers represents surplus of workers that will bring about a downward pressure on the wage rate.

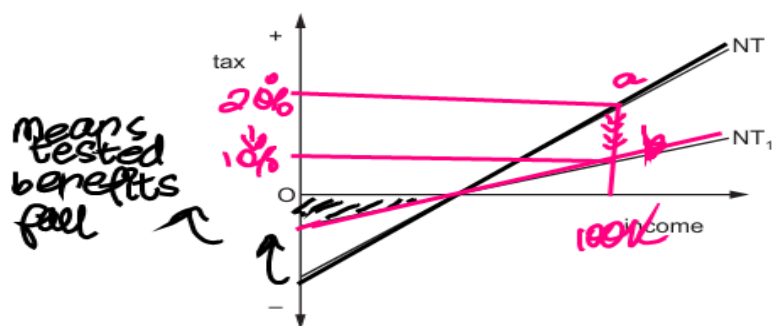
C – An increase in the number of people unemployed represents surplus of workers that will bring about a downward pressure on the wage rate.

D – An increase in labor productivity will increase the marginal revenue productivity or the demand for labor that will bring about an upward pressure on the wage rate.

15-A:

15 A country has a negative income tax regime.

The curve NT in the diagram shows the country's initial tax schedule.



A change in the tax rate causes the schedule to shift to NT₁.

How will this affect work incentives and the after-tax distribution of income?

	work incentives	distribution of income
A	strengthen ✓	less equal ✓
B	strengthen ✓	more equal
C	weaken	less equal
D	weaken	more equal

It can be seen in the diagram above that means tested benefits are falling, and also, the rate of taxation is also seen to fall for high income groups.

As a result of this, the distribution of income will become less equal.

However since the rate of taxation is falling, this indicates that work incentives will rise, as workers will be able to keep more of the additional income they earn. This explains why option A is correct.

16-D:

A – Allowing wage rates to fall will not help the firm to meet the rising demand for labor when the supply of labor is fixed, or perfectly inelastic.

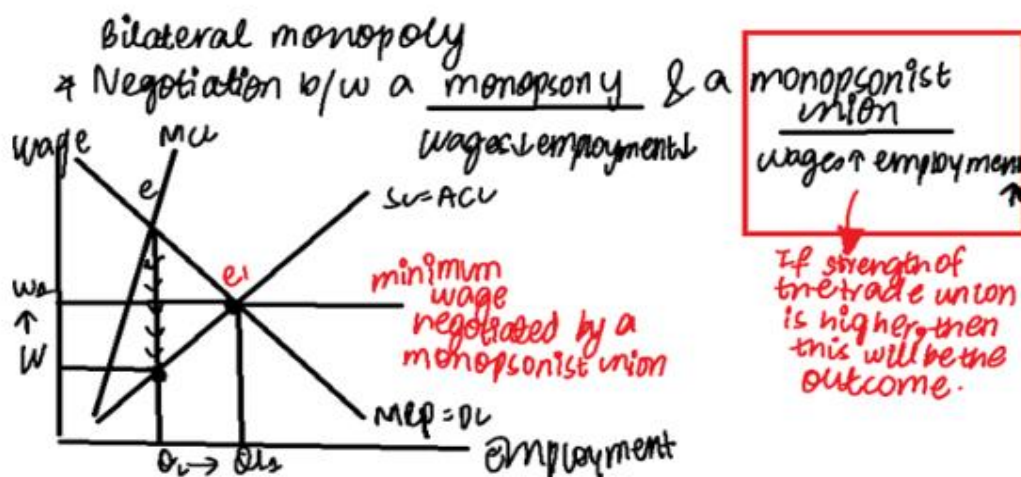
B , C: Introducing work sharing and reducing working hours(option B) and making offices more attractive by adding air conditioning(option C) will benefit current workers, and does not increase the fixed supply of skilled workers.

D – Providing more training courses for new recruits will increase the supply of skilled labor, thus making this option correct .

17-B:

Trade union power (strengthens):

When the trade union power strengthens, possibly because of it being the largest supplier of workers, making it a monopsonist union, it is most likely to negotiate an improvement in wages and employment prospects at the same time.



Immigration controls (strengthen): If the immigration controls are strengthened, this means that there will be a less availability of labor, and so, due to a fall in supply of labor, there will be a rise in wages paid, and since there would now not be a surplus of labor, this will improve the employment prospects too.

National minimum wage raised: As can be seen in the diagram above, if the national minimum wage is raised to the equilibrium by a monopsonist union, this will result in a rise in wages paid, and also, improve the employment prospects too.

This explains why key B is correct.

18-B:

A - Imperfect information is a source of market failure.

-> If individuals are not aware of the benefits or positive externalities associated with consuming merit goods, they may not prioritize their consumption, even if it is in their best interest or contributes to the overall welfare of society. For example, preventive healthcare services or education may be underutilized if individuals are not fully informed about their long-term benefits. This under-allocation of merit goods results in market failure.

-> If individuals are unaware of the harmful consequences or negative externalities associated with demerit goods, they may over- consume them without fully considering the long-term repercussions. For example, individuals may engage in excessive consumption of harmful substances such as: alcohol, due to a lack of information about the health risks involved. This over -allocation of de-merit goods results in market failure.

B) Income Inequality: Income inequality refers to the unequal distribution of income among individuals or households in an economy. This is a macro-economic concern and may not cause market failure. This is why, key B, becomes correct.

C) Monopoly is a source of market failure. A “monopoly” occurs when a single firm dominates the market and restricts competition. In a monopoly, the monopolistic firm has significant market power, allowing it to set prices higher than in a competitive market, resulting in reduced output and consumer welfare. The under-allocation of output and the existence of deadweight loss results in “market failure”.

D:) Non-excludability is a source of market failure, particularly concerning public goods. Public goods are goods or services that are non-excludable, meaning that payment by one individual does not restrict others from enjoying the benefits associated with these goods, even the non-tax payers get to consume it. This results in the free-rider problem. Due to the free-rider problem, where individuals can benefit from public goods without paying for them, private markets do not provide for them, and this non-provision of public goods, results in market failure.

19-B:

A – If government intervention in form of subsidies results in “cheaper fresh fruit and vegetables”, this represents a success of the policy.

B – When an “increase in production” of fresh fruits and vegetables leads to unintended consequences such as: land clearance reducing wildlife, this represents a failure of the policy. This is why, key B, becomes correct.

C – If government intervention in form of subsidies, results in “lower spending in treating dietary illnesses”, this again represents a success of the policy.

D - If government intervention in form of subsidies, results in creation of more jobs picking fruits and vegetables, this again represents a success of the policy.

A – Cyclical unemployment is also known as demand deficient unemployment which is associated with a deficiency in Aggregate demand , or in other words, too low of Aggregate demand.

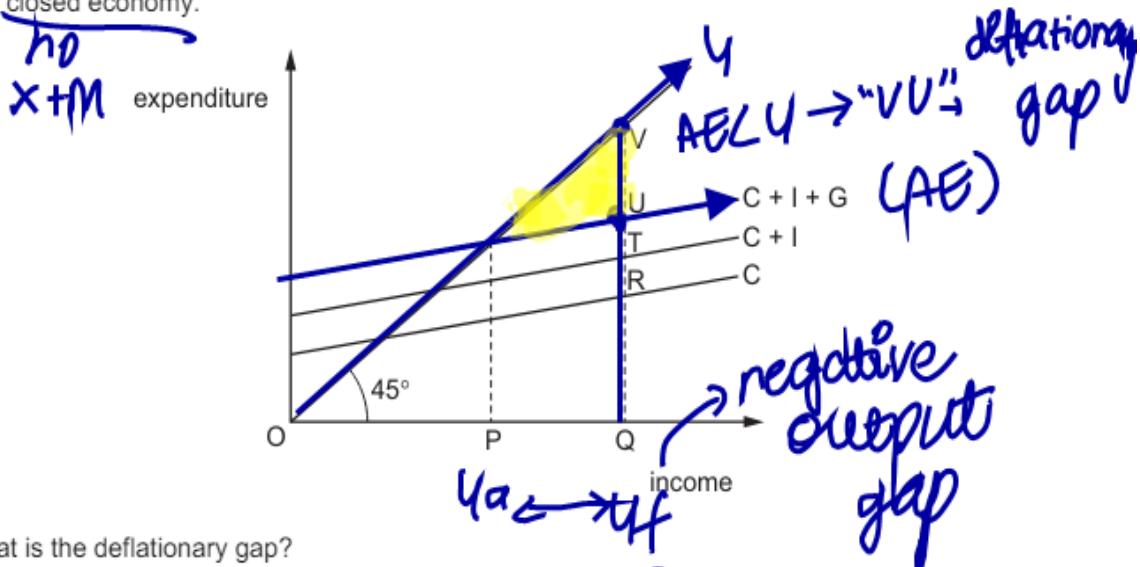
C – Structural unemployment arises when the structure of the economy changes. This occurs when there is a change in the pattern of demand and supply of the economy.

D – Voluntary unemployment occurs when workers are unwilling to work at the current wage rate. This is because, there is a lesser gap between unemployment benefits and the take home pay of the employed.

A deflationary gap is an economic concept that refers to a situation where the aggregate expenditure of an economy is below its potential level of national income leading to a gap in the overall level of economic activity.

It is measured by how much the actual aggregate expenditure falls short of potential level of national income at the full employment level of income.

21 In the diagram, OP is the equilibrium level of income and OQ the full employment level of income in a closed economy.



What is the deflationary gap?

- A** PQ **B** RV **C** TV **D** UV

22-B:

The correct answer is B) interest payments on government bonds.

Injectors into the circular flow of income are expenditures that enter the economy which boost aggregate demand.

- ➔ Interest payments on government bonds **is an example of government expenditure** and is considered an injection because they represent a transfer of money from the government to bondholders, who can then spend or invest that money, stimulating economic activity.

A) Corporate taxes are a leakage from the circular flow.

C) Payment of dividends to foreign shareholders is also a leakage, as it represents a transfer of money out of the country.

D) Repayment of bank loans will result in less funds available for re-investment as a result of which, injections will reduce.

23-A:

A - The dependency ratio is the number of dependents (children and elderly) relative to the working-age population. An increase in the dependency ratio means that there are more dependents relying on each working-age individual, which can lead to reduced household savings and investment as a result of which, there will be a fall in the rate of economic growth that will result in a decrease in GDP per head.

B) Labor productivity might not decrease as a result of an increase in dependency ratio as it depends on various factors like technology, education, and innovation.

C) An increase in the dependency ratio is unlikely to lead to an increase in the labor force, as dependents are not part of the workforce.

D) Unemployment might not increase because of a rise in the dependency ratio. This is because, the dependents are not a part of working age population.

24-C:

Economic growth refers to an increase in a country's production capacity, measured by an increase in GDP. Economic development, on the other hand, refers to a broader process of improvement in the quality of life, including social, economic, and political progress.

A) Economic development is not necessary for economic growth, as growth can occur without necessarily leading to development.

B) Economic growth and economic development are not directly proportional, as growth can occur without development.

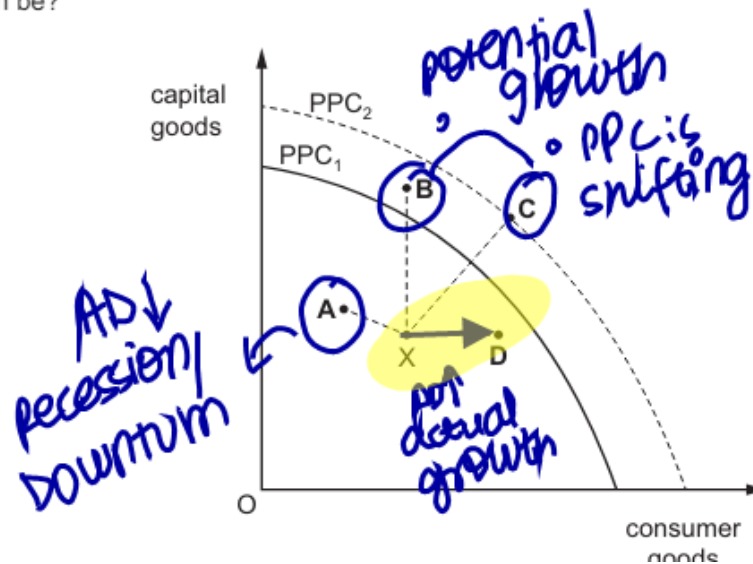
C) This is correct because economic growth provides the resources and opportunities for economic development and **may enable it. Because of a economic growth, there will be funds available with the government for investments in human capital, infrastructure, institutions, and technological progress, among other factors, which can lead to economic development.**

D) Economic growth is not always sustainable, as it can be driven by unsustainable factors like resource depletion or environmental degradation.

25-D:

25 The diagram shows a production possibility curve PPC_1 . The economy is initially at point X.

If the economy achieves actual economic growth but not potential growth, what would the final position be?



The diagram above shows that since the economy is achieving actual growth due to a rise in Aggregate demand, actual that is the point inside the PPC shown through "X" will move closer to the potential, shown through "point D"

26-B:

A) Political instability can discourage foreign direct investment, leading to a shortage of capital and an increased need of borrowing.

B) Government subsidies for tourism development are unlikely to contribute to this, as they are aimed at promoting economic growth and development, not increasing debt.

C) Withdrawal of multinational companies due to delays in government contracts can lead also lead to reduced foreign exchange earnings and increased borrowing.

D) **Terms of Trade (TOT):** The ratio of export prices to import prices, expressed as:

$$TOT = (\text{Index of export prices} / \text{Index of import prices}) * 100$$

Worsening TOT: A decrease in the TOT ratio, meaning:

- Export prices fall relative to import prices
- A country gets fewer imports for each unit of exports

In the context of developing countries, a worsening TOT with partners in developed economies means they face unfavorable trade conditions, leading to reduced export earnings and increased import costs. This can contribute to increased external debt, as they may need to borrow to finance their trade deficits.

27-A:

The "infant industry" argument suggests that developing countries should protect their young industries from foreign competition to allow them to grow and develop. However, the condition of allowing free trade (Option A) would conflict with this argument, as it would require the country to open up its markets to foreign competition, potentially harming its developing industries.

The other options do not directly conflict with the "infant industry" argument:

B) Controlling inflation is a macroeconomic goal that does not directly relate to trade policy or industry development.

C) Contractionary fiscal policy aims to reduce government spending to decrease the size of budget deficits, which may not harm infant industries.

D) Privatizing government enterprises can lead to efficiency gains, but does not directly impact trade policy or industry protection.

28-A:

Options B and C are similar in a sense that both of them represent a fixed exchange rate. With a fixed exchange rate, a central bank is limited in its ability to adjust interest rates, because it has to maintain the value of the domestic currency. Let's say the central bank has chosen to artificially revalue its currency, now in face of deflation, it will not have autonomy to lower its interest rates, because it will lead to an immediate impact on the exchange rate, as its value can fall because of hot money outflows. This is why, this is incorrect.

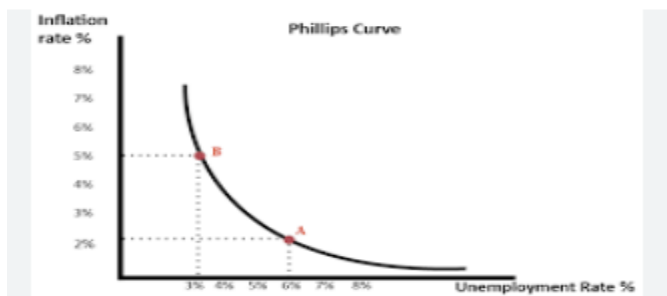
A - Therefore, that being said, it will only be in a floating exchange rate regime(option D), that a central bank can exert close control over its monetary policy, since it has an autonomy and more flexibility to adjust interest rates, thus making it correct.

In a monetary union (option D), the central bank cannot influence the interest rate of a particular country, this is because, a single interest rate is set for all of the member countries. This is because, in such a union, all countries share the same monetary policy, an example of which, is the Eurozone. This is why, this is incorrect too.

29-C:

On the vertical axis of Philips curve, rate of inflation is measured. Whereas, on the horizontal axis of the Philips curve, the rate of unemployment is written.

This explains why only key C will be correct.



30-C:

C - The productivity gap between workers in poor and rich countries is often due to differences in human capital, such as education and skills. Investing in education in poor countries would help to improve worker skills and knowledge, which is most likely to increase labor productivity.

A) Increasing the saving ratio in poor countries could lead to more investment, but may not necessarily improve worker productivity.

B) Freedom of migration could lead to brain drain from poor countries, exacerbating the productivity gap.

D) Removing trade barriers could increase trade, but may not directly impact worker productivity.

